Can Frontier Agents Invent Robot Tools?

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Abstract: We study whether Opus 5.5, Astra 6, and Sol 6.1 can design tools for five manipulation tasks using a simulated Franka Panda. Each model proposes a tool and a sequence of robot movements, tests them in simulation, and revises its proposal over four attempts. Averaged across the five tasks, success rates are 90.7% for Astra 6, 86.7% for Sol 6.1, 73.3% for Opus 5.5.

Code and data: https://mateig.github.io/agent-tool-design

Keywords: Agentic Robotics, Tool Design

1 Introduction

A custom tool can give a robot the contact geometry needed for a task, such as a thin blade for lifting a plate or a prop for holding a door open. Designing it requires choosing both its shape and movements that the robot can execute.

Prior work studies tool shape and control jointly [1]. RobotSmith [2] and VLMgineer [3] use language models to generate tools for manipulation tasks. We study the reliability of this approach by comparing three frontier models on the same five tasks, with four attempts to design each tool and its movements.

2 Method

We evaluate five tasks using a Franka Panda in MuJoCo [4] (Table 1). Each model receives scene dimensions and two images, then specifies a tool of up to 32 rigid primitives and up to 60 gripper poses over 90 seconds. The tool starts attached to the gripper, and an inverse-kinematics controller executes the proposed poses.

Task	What the robot must do
Tunnel	Pull a block out through a low opening.
Plate	Slide under a plate and hold it above the table.
Knob	Reach into a cavity and turn a square knob.
Buttons	Press two separated, guarded buttons at the same time.
Door	Leave a prop holding a spring-loaded door open, then retrieve a block.

Table 1: Five manipulation tasks.

For each task, a model proposes a tool and its movements, tests them in three training scenes, and receives scores and rollout images. It can revise its proposal over four attempts, each limited to 16,000 output tokens. Incomplete responses count as failed attempts. We repeat this process independently three times per model and task, for 45 design runs in total. Within each run, we select the attempt with the most training successes, breaking ties by validity, task progress, then attempt order. We test the selected tool and movements in ten slightly shifted scenes.

3 Results

Model	Tunnel	Plate	Knob	Buttons	Door	Mean
Astra 6	**100.0**	**100.0**	86.7	**100.0**	**66.7**	**90.7**
Sol 6.1	**100.0**	**100.0**	**100.0**	**100.0**	33.3	86.7
Opus 5.5	**100.0**	**100.0**	66.7	**100.0**	0.0	73.3

Table 2: Task success rates (%) over 30 trials per model and task.

Across the five tasks, the models achieve an overall success rate of 83.6% over 450 trials. All three models achieve 100% success on tunnel retrieval, plate lifting, and simultaneous button pressing (Table 2, Figure 1). Sol performs best on the recessed knob, while Astra performs best on door propping.

Selecting among four attempts increases the success rate from 70.0% for the first attempts to 83.6%.

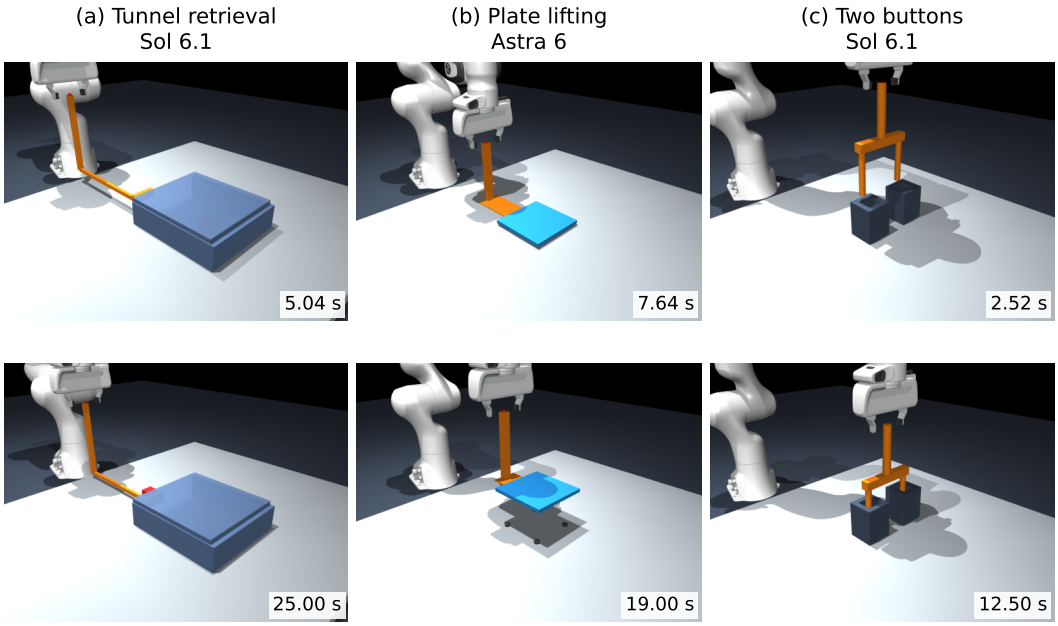


Figure 1: Successful rollouts for tunnel retrieval (Sol 6.1), plate lifting (Astra 6), and simultaneous button pressing (Sol 6.1). Each column shows an intermediate frame above the completed task. Agent-designed tools are orange.

4 Conclusion

Frontier models can design tool shapes and robot movements that complete several manipulation tasks in simulation. Selecting among four attempts raises overall success from 70.0% to 83.6%, supporting the use of repeated design attempts with simulation feedback.

References

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